- $\Bbbk = \mathbb{Q}(q,t)$
- $F = \Bbbk[c_1^{\pm 1}, c_2^{\pm 1}]$
- $\Lambda_F(X^{m,n}) \cong \Lambda_F(X)$ for $\gcd(m,n)=1$
- Λ : algebra of symmetric functions
- Hopf algebra structure of Λ (Use formal alphabets X, Y to distinguish $\Lambda \otimes \Lambda \cong \Lambda(X)\Lambda(Y)$)
  - coproduct : $\Delta f := f[X+Y]$ ($\Delta : \Lambda \to \Lambda \otimes \Lambda$: $\Bbbk$ algebra homomorphism)
  - antipode : $S(f) := f[-X]$ ($S : \Lambda \to \Lambda$: automorphism)

- For any algebra A containing a copy of Λ, there is an adjoint action of $f \in \Lambda$ on $\zeta \in A$ is

$$(\mathrm{Ad}\, f)\zeta = \sum_i f_i \zeta g_i \quad \text{where } f[X-Y] = \sum_i f_i(X) g_i(Y)$$

$(1 \otimes S)\Delta f = \sum_i f_i \otimes g_i$

This is why we write $f[X-Y]$ instead of $f[X+Y]$

Let $(A, \Delta, \varepsilon, S)$ be a Hopf algebra
Recall the adjoint representation of A on itself as:
$\forall a, b \in A$

$$a \cdot b = (\mathrm{ad}\, a) b := \sum_i a_i b S(a_i') \quad \text{where } \Delta(a) = \sum_i a_i \otimes a_i'$$

* A is an $(A \otimes A)$-module via $(a \otimes a') \cdot b = a b S(a')$ $\forall a, a', b \in A$
* This is just making A into an A-module via Δ.

e.g. $p_n[X+Y] = p_n(X) + p_n(Y) = p_n(X) \cdot 1 + 1 \cdot p_n(Y)$

$\therefore (\mathrm{Ad}\, p_n)\zeta = p_n(X)\zeta \cdot 1 + 1 \cdot \zeta p_n[-Y]$

$= p_n(X)\zeta - \zeta p_n(Y) = [p_n, \zeta]$.

e.g. $\Omega[zX + zY] = \Omega[zX]\Omega[zY]$

$\therefore (\mathrm{Ad}\, \Omega[zX])\zeta = \Omega[zX]\zeta\Omega[-zX]$

e.g. $h_n[X+Y] = \sum_{j+k=n} h_j(X) e_j(Y)$

$\therefore (\mathrm{Ad}\, h_n)\zeta = \sum_{j+k=n} h_j(X)\zeta e_j[-Y] = \sum_{j+k=n} (-1)^k h_j \zeta e_k$

- Schiffmann Algebra $\mathcal{E}$ is generated by a central Laurent polynomial subalgebra F ($\Bbbk[c_1^{\pm 1}, c_2^{\pm 1}]$) and $\{\Lambda_F(X^{m,n}) : (m,n) = 1\}$ (family of subalgebras) satisfying 3 defining relations

↳ Read Extended Delta Conjecture §3.1 - 4.1